A reproducible computational framework for decoupled wildfire ignition and propagation benchmarking

D2 targets | M3/M2 inference | open benchmark against Canadian FWI and INPE

INPUTS

- **VIIRS active fires**
- Suomi-NPP, NOAA-20/21 375-m detections
- **ERA5-Land weather**
- VPD, rain, wind, FFMC/ISI
- **MapBiomas C2 fuel**
- Pasture, savanna, forest
- **Causal fire history**

D2 TARGET OPERATOR

Primary ignition

- fire at cell i, day t
- no t-1 fire within 25 km
- no same-cell fire in 10 d

Tactical propagation

- fire at cell i, day t
- within 25 km of a t-1 fire

The two target domains are disjoint.

D2 is applied once and reused for training, test and alerts.

BIOPHYSICAL GATE

Applied before inference

- **fuel > 0**
- **VPD ≥ 0.8 kPa**
- **48-h rain < 10 mm**

Removes water, urban surfaces and saturated fuels from inference.

The gate is not a label filter.

DECOUPLED INFERENCE

M3 hybrid ignition

weather + fuel + ecotone
+ anthropogenic priors
no t-1 fire input

M2 fallback

weather + fuel only
used where fire history
is absent

Propagation

causal t-1 kernel:
distance, counts and FRP

Decoupled, not persistence.

EVALUATION + OUTPUTS

- Temporal test 2024–2025
- 50-km spatial blocks
- Grouped bootstrap
- Lead-time analysis
- Cost-loss and conformal

Alert maps
Benchmark tables
Open code and data

Every result is tied to a frozen split.

Contribution: an auditable target operator, gated inference and evaluation harness — not a new boosting algorithm.